Kanishk Devatwal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.14
Royal International School	Class XII (RBSE)	2023	81.80%
Royal International School	Class X (RBSE)	2021	96.00%

WORK EXPERIENCE

The Bank of New York (BNY) -- Software Engineering Intern

- Built an **AI-powered Code Lineage Agent** on the Eliza platform that analyzed Java Spring Boot projects and traced how data moves from **Controller Service DAO Database**. Used Python AST parsing to generate SQL-to-report-field mappings and integrated GitLab APIs and MCP-based file retrieval to answer questions about the codebase.
- Improved the code analysis process by replacing full repository downloads with on-demand file retrieval, reducing analysis time from **30+ minutes to around 30 seconds (~95% improvement)**.
- Built an **AI-powered Holiday Validation Agent** on the Eliza platform to automate verification of **holiday dates and market closing information** from bank emails, validating extracted details against a **95+ country dataset** using GraphQL.

SKILLS

Languages: C, C++, Python, JavaScript

Libraries/Frameworks: Pandas, Matplotlib, SciPy, Flask

APIs & Technologies: GraphQL, GitLab REST API, MCP, AST Parsing (Javalang)

Developer Tools: Git, VS Code, Jupyter Notebook, Google Colab, Termux

Others: Verilog, MATLAB, Fusion 360, LaTeX, HTML/CSS

RELEVANT COURSES

Electrical Engineering: Digital Signal Processing, Signals and Systems, Introduction to CMOS Technology, Semiconductor Device Fundamentals, Analog Electronics, Digital Systems, Electrical Machines, Analog Circuits, Power Electronics, Error Correcting Codes

Mathematics: Probability and Random Processes, Matrix Theory, Vector Calculus, Differential Equations, Complex Variables, Calculus-1, Calculus-2

Computer Science: Data Structures and Applications, Intro to Programming, Artificial Intelligence, Information Theory and Coding, Computer Architecture, Communication Systems

Labs: Electronic Devices and Circuits Lab, Digital Signal Processing Lab, Electrical Circuits Lab, Digital Fabrication, Electrical Machines Lab

PROJECTS

64-Point FFT Implementation Using Radix-2^2 SDF Architecture -- GitHub

- Designed and implemented a **fully pipelined 64-point FFT processor** using a **Radix-2^2 Single Path Delay Feedback (SDF)** architecture in Verilog HDL, enabling continuous streaming with one sample processed per clock cycle.
- Implemented the FFT using **butterfly units, delay buffers, twiddle-factor ROM, and complex multipliers** with Q15 fixed-point arithmetic, reducing multiplier and memory requirements compared with a conventional Radix-2 architecture.
- Verified the design against a **MATLAB fixed-point reference model** across multiple test cases, achieving **less than 1 LSB maximum error** and correctly identifying frequency peaks in the bit-reversed output stream.

- Synthesized and deployed the design on an **Intel Cyclone V FPGA (DE10-Standard)**, achieving a **181.23 MHz maximum operating frequency** with **71-cycle pipeline latency**; used **SignalTap** for hardware-level pipeline and data-stream debugging.

Adaptive Noise Cancellation in Headphones -- GitHub

- Designed a dual-mode real-time adaptive noise cancellation system using **MATLAB**.
- Implemented a **Normalized Least Mean Squares (NLMS)** filter with order M=8 and step size =0.008 to estimate and cancel background noise using a reference signal.
- Added a programmable **notch filter** centered at 1000 Hz with pole radius r=0.998 to remove dominant tonal noise before applying NLMS with M=12 and =0.00008.
- Achieved up to **26.56 dB SNR improvement** during full suppression, demonstrating effective noise reduction while preserving speech.
- Chose **NLMS over LMS** for improved stability with fluctuating input power and lower computational cost compared with RLS.

Home Devices Management System - C++ GitHub

- Developed a Home Device Management System to address daily smart home automation needs, enabling secure device control for multiple user roles.
- Implemented robust backend logic in **C++**, applying Object-Oriented Programming principles such as **abstraction, inheritance, and polymorphism**.
- Designed a modern desktop interface using **Electron, HTML, CSS, and JavaScript** for intuitive user interaction and authentication.

EXTRACURRICULAR

- Participated in various social initiatives including Plantation Drives, Swachh Bharat campaigns, awareness walkathons, essay writing, Tidy Life, and other community activities, contributing 60+ hours as a volunteer.

- Actively involved in sports and fitness, with interests in Volleyball and Gym training.

PROGRAMMING ACHIEVEMENTS

Solved **500+ problems** across all coding platforms. LeetCode | Codeforces